```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
!unzip /content/drive/MyDrive/pothole_dataset.zip -d /content/
```

```
Archive:  /content/drive/MyDrive/pothole_dataset.zip
  inflating: /content/README.roboflow.txt
  inflating: /content/data.yaml
   creating: /content/test/
   creating: /content/test/images/
 extracting: /content/test/images/potholes100_png.rf.d0a83212b1fb944f869fb41f85bcc413.jpg
 extracting: /content/test/images/potholes106_png.rf.1d110c6aa5cd94fd4950658d746ff463.jpg
 extracting: /content/test/images/potholes107_png.rf.c7683df14b1e90bb56ca5e1854186cbe.jpg
 extracting: /content/test/images/potholes10_png.rf.5f3b635f90f42473a9dcfae67b462090.jpg
 extracting: /content/test/images/potholes111_png.rf.9a0479ea059d2ae13292334a031ce03a.jpg
 extracting: /content/test/images/potholes113_png.rf.5051ef1f136397a52a3fc604898bcea8.jpg
 extracting: /content/test/images/potholes129_png.rf.0ae0bf6aa390fe747638e0960b88f71d.jpg
 extracting: /content/test/images/potholes142_png.rf.a25a03fafabb963b6dc772870e5df5bb.jpg
 extracting: /content/test/images/potholes145_png.rf.165126ad88fadc15c0b5caedc5489090.jpg
 extracting: /content/test/images/potholes148_png.rf.f14865f99f5a23e3723ab315639fbb70.jpg
 extracting: /content/test/images/potholes151_png.rf.c258aa0b8bfbf3251cf6684b26d4f325.jpg
 extracting: /content/test/images/potholes154_png.rf.9f85eae652c46b6c7d6b671d7f2b0804.jpg
 extracting: /content/test/images/potholes158_png.rf.b2991cf6863ea38cf94daeead3032d04.jpg
 extracting: /content/test/images/potholes185_png.rf.a18c474199ae2ea71ec8b54a1bedd0d2.jpg
 extracting: /content/test/images/potholes197_png.rf.d8cf88cd3bb660725782abd2dc94556b.jpg
 extracting: /content/test/images/potholes200_png.rf.37c5c5e7e40303d97021c0764c12abd5.jpg
 extracting: /content/test/images/potholes227_png.rf.f95067ca71bfb14e84ae56f154ef052a.jpg
 extracting: /content/test/images/potholes22_png.rf.d0b0341089552ce9171ab7957c5623f6.jpg
 extracting: /content/test/images/potholes230_png.rf.ecb66a9b99f678c208d3591bfd857e44.jpg
 extracting: /content/test/images/potholes243_png.rf.d35f2ff56f7a311100f24d88be6d2123.jpg
 extracting: /content/test/images/potholes245_png.rf.e2dbe4ba4dc90b3424a9d5b2391b8d2a.jpg
 extracting: /content/test/images/potholes262_png.rf.7487bef3803026b8900aedda32da3043.jpg
 extracting: /content/test/images/potholes275_png.rf.804630de6c1a2884604e5d87d5eb0ca0.jpg
 extracting: /content/test/images/potholes283_png.rf.17094e0305fc42e33533103e7dcbdaa5.jpg
 extracting: /content/test/images/potholes290_png.rf.399db13e76f745660c912e76a0092bed.jpg
 extracting: /content/test/images/potholes296_png.rf.35ed863329afca114c2db6a41238ffa6.jpg
 extracting: /content/test/images/potholes2_png.rf.bf7891a70a2a2b1d9038198f55de6d62.jpg
 extracting: /content/test/images/potholes313_png.rf.268e443a3c33545c5bc1a43470e619aa.jpg
 extracting: /content/test/images/potholes322_png.rf.81e60c2eef0d1ffc69fbb400302bea19.jpg
 extracting: /content/test/images/potholes324_png.rf.67009da534f599be367040016322d54e.jpg
 extracting: /content/test/images/potholes347_png.rf.8cdebc5fb1d491cf6b227a660a0e9ed1.jpg
 extracting: /content/test/images/potholes34_png.rf.5e031debd3003ee0ef8ead2757e8b3b2.jpg
 extracting: /content/test/images/potholes36_png.rf.c5d17f487745b7d61c4b66cf5cb8f116.jpg
 extracting: /content/test/images/potholes37_png.rf.144e4d401f783923032bd91cd5ed15c8.jpg
 extracting: /content/test/images/potholes383_png.rf.5e4fa2e58a24f3cf36684eccc5b1960e.jpg
 extracting: /content/test/images/potholes398_png.rf.14e64af7c32ca31837933ad9b6e83f49.jpg
 extracting: /content/test/images/potholes404_png.rf.f3b237b0d3133adc9d45c71954f8cdd5.jpg
 extracting: /content/test/images/potholes421_png.rf.c26261c0e376b041eedaf33b5815d794.jpg
 extracting: /content/test/images/potholes426_png.rf.6b1fe882b9ce0d7a9c7d8f4f81df5a32.jpg
 extracting: /content/test/images/potholes428_png.rf.3dba37e44e424ef3df7e339447855a0f.jpg
 extracting: /content/test/images/potholes42_png.rf.48e6dd9f72cc0675a047075d4fc7a527.jpg
 extracting: /content/test/images/potholes436_png.rf.589f233a13603a57d593a6c2141562bb.jpg
 extracting: /content/test/images/potholes446_png.rf.62e3d54f13fad7ccac0af1df4bd79758.jpg
 extracting: /content/test/images/potholes459_png.rf.7fe48433a565afb2c868097b51782246.jpg
 extracting: /content/test/images/potholes466_png.rf.8c4b0a1b91979b92a0d8bce2ff4d66b9.jpg
 extracting: /content/test/images/potholes476_png.rf.961310b4bdf215ba401d77b356ffc6d6.jpg
 extracting: /content/test/images/potholes484_png.rf.c13862ed38094f3f19d2d993e780cb73.jpg
 extracting: /content/test/images/potholes496_png.rf.51db5aa2f9d6c6ef3194a0ad403b06e8.jpg
 extracting: /content/test/images/potholes506_png.rf.b009008ef5cae12eec85bee8087cd17d.jpg
 extracting: /content/test/images/potholes507_png.rf.1c11800f4b218a0fe249e3b44a494d51.jpg
 extracting: /content/test/images/potholes50_png.rf.cf5bfd656790b1794de36e69580731dd.jpg
 extracting: /content/test/images/potholes513_png.rf.71907ed6742e664a90cbccfce999e162.jpg
 extracting: /content/test/images/potholes531_png.rf.96e5a52b1ba8cf312b4bb96d83a67d89.jpg
```

```
!pip install ultralytics
```

```
Collecting ultralytics
  Downloading ultralytics-8.4.17-py3-none-any.whl.metadata (39 kB)
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (4.13.0.92)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.10.0+cpu)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.25.0+cpu)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)
```

```
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.31.0)
Collecting ultralytics-thop>=2.0.18 (from ultralytics)
  Downloading ultralytics_thop-2.0.18-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralyt
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultra
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.1
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.24.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralyt
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.14.0
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.6.
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2025.3
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.8.0->
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.8.0->ultralytic
Downloading ultralytics-8.4.17-py3-none-any.whl (1.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2/1.2 MB 13.5 MB/s eta 0:00:00
Downloading ultralytics_thop-2.0.18-py3-none-any.whl (28 kB)
Installing collected packages: ultralytics-thop, ultralytics
Successfully installed ultralytics-8.4.17 ultralytics-thop-2.0.18
```

```
from ultralytics import YOLO
```

```
Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics
```

```
!ls /content/
```

```
data.yaml  drive  README.roboflow.txt  sample_data  test  train  valid
```

```
!cat /content/data.yaml
```

```
train: ../train/images
val: ../valid/images
test: ../test/images

nc: 1
names: ['pothole']

roboflow:
  workspace: vishus
  project: pothole-detection-gfdyz
  version: 1
  license: Private
  url: https://app.roboflow.com/vishus/pothole-detection-gfdyz/1
```

```
%%writefile /content/data.yaml
path: /content
train: train/images
val: valid/images
test: test/images

names:
  0: pothole
```

```
Overwriting /content/data.yaml
```

```
!ls /content/train/images | head
!ls /content/train/labels | head
```

```
potholes0_png.rf.b2f5d84664f023ee15907014f4da3814.jpg
potholes101_png.rf.fdd2ba5cbf0c7b30d46f02111ac4b609.jpg
potholes102_png.rf.f46cd0ad285017b45bfb134f8348a8da.jpg
potholes103_png.rf.5c8a2116eb67c72ce325d2f3f378a11d.jpg
potholes104_png.rf.ab6d1a201065d7f79b8a2053af83ee84.jpg
potholes108_png.rf.77b65d635f30f3e8664f8036bc2cd69d.jpg
potholes109_png.rf.23ba52fc66f4bed802b863adb93a6856.jpg
potholes110_png.rf.d29a6ec5c5ee1cf2e43308e3546a9468.jpg
```

```
potholes112_png.rf.85a896d3f47cfe99f3a71d53c8f02043.jpg
potholes114_png.rf.e837f9cb49cf54be8c293c7ec7ed1e58.jpg
potholes0_png.rf.b2f5d84664f023ee15907014f4da3814.txt
potholes101_png.rf.fdd2ba5cbf0c7b30d46f02111ac4b609.txt
potholes102_png.rf.f46cd0ad285017b45bfb134f8348a8da.txt
potholes103_png.rf.5c8a2116eb67c72ce325d2f3f378a11d.txt
potholes104_png.rf.ab6d1a201065d7f79b8a2053af83ee84.txt
potholes108_png.rf.77b65d635f30f3e8664f8036bc2cd69d.txt
potholes109_png.rf.23ba52fc66f4bed802b863adb93a6856.txt
potholes110_png.rf.d29a6ec5c5ee1cf2e43308e3546a9468.txt
potholes112_png.rf.85a896d3f47cfe99f3a71d53c8f02043.txt
potholes114_png.rf.e837f9cb49cf54be8c293c7ec7ed1e58.txt
```

```
# from ultralytics import YOLO

# model = YOLO("yolov8n.pt")

# model.train(
#     data="/content/data.yaml",
#     epochs=50,
#     imgsz=640,
#     batch=16
# )

# model.train(
#     data="/content/data.yaml",
#     epochs=50,
#     imgsz=640,
#     batch=16,
#     project="/content/drive/MyDrive/pothole_yolo",
#     name="train1",
#     exist_ok=True
# )
```

```
from ultralytics import YOLO

MODEL_PATH = "/content/drive/MyDrive/pothole_yolo/train1/weights/last.pt"

import os

if os.path.exists(MODEL_PATH):
    print("Resuming previous training...")
    model = YOLO(MODEL_PATH)
    model.train(resume=True)
else:
    print("Starting fresh training...")
    model = YOLO("yolov8n.pt")
    model.train(
        data="/content/data.yaml",
        epochs=100,
        imgsz=640,
        batch=16,
        project="/content/drive/MyDrive/pothole_yolo",
        name="train1",
        exist_ok=True
    )
```

```
Resuming previous training...
Ultralytics 8.4.17 🚀 Python-3.12.12 torch-2.10.0+cpu CPU (AMD EPYC 7B12)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, c
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1KB 15.2MI

                   from  n    params  module                                        arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv              [3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv              [16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f              [32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv              [32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f              [64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv              [64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f              [128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv              [128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f              [256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF             [256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample          [None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat            [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f              [384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample          [None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat            [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f              [192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv              [64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat            [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f              [192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv              [128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat            [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f              [384, 256, 1]
 22        [15, 18, 21]  1    751507  ultralytics.nn.modules.head.Detect            [1, 16, None, [64, 128, 256]]
Model summary: 130 layers, 3,011,043 parameters, 3,011,027 gradients, 8.2 GFLOPs

Transferred 355/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
train: Fast image access ✅ (ping: 0.0±0.0 ms, read: 2379.8±986.0 MB/s, size: 61.9 KB)
train: Scanning /content/train/labels... 465 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 465/465 2.4Kit/s 0.2s
train: New cache created: /content/train/labels.cache
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_averag
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1829.4±589.2 MB/s, size: 65.6 KB)
val: Scanning /content/valid/labels... 133 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 133/133 2.6Kit/s 0.1s
val: New cache created: /content/valid/labels.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'moment
optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0
Plotting labels to /content/drive/MyDrive/pothole_yolo/train1/labels.jpg...
---------------------------------------------------------------------------
AssertionError                            Traceback (most recent call last)
/tmp/ipython-input-299/2157348755.py in <cell line: 0>()
      8     print("Resuming previous training...")
      9     model = YOLO(MODEL_PATH)
---> 10     model.train(resume=True)
     11 else:
     12     print("Starting fresh training...")

                              4 frames
/usr/local/lib/python3.12/dist-packages/ultralytics/engine/trainer.py in resume_training(self, ckpt)
    933             return
    934         start_epoch = ckpt.get("epoch", -1) + 1
--> 935         assert start_epoch > 0, (
    936             f"{self.args.model} training to {self.epochs} epochs is finished, nothing to resume.\n"
    937             f"Start a new training without resuming, i.e. 'yolo train model={self.args.model}'"

AssertionError: /content/drive/MyDrive/pothole_yolo/train1/weights/last.pt training to 100 epochs is finished, nothing to
resume.
Start a new training without resuming, i.e. 'yolo train model=/content/drive/MyDrive/pothole_yolo/train1/weights/last.pt'
```

```
BEST_MODEL = "/content/drive/MyDrive/pothole_yolo/train1/weights/best.pt"
```

```
model = YOLO(BEST_MODEL)

metrics=model.val(data='/content/data.yaml')
print(metrics)
```

```
Ultralytics 8.4.17 🚀 Python-3.12.12 torch-2.10.0+cpu CPU (AMD EPYC 7B12)
Model summary (fused): 73 layers, 3,005,843 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 2215.3±336.5 MB/s, size: 68.8 KB)
val: Scanning /content/valid/labels.cache... 133 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 133/133 34.9Mit/s 0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 9/9 3.0s/it 27.4s
                   all        133        411      0.865      0.652       0.81      0.521
Speed: 4.3ms preprocess, 195.1ms inference, 0.0ms loss, 0.8ms postprocess per image
Results saved to /content/runs/detect/val
ultralytics.utils.metrics.DetMetrics object with attributes:
```

```
ap_class_index: array([0])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at 0x7d557f8404d0>
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-Confidence(B)', 'Recall-Confidence(B)']
curves_results: [[array([          0,    0.001001,    0.002002,    0.003003,    0.004004,    0.005005,    0.006006,    0.00700
          0.024024,    0.025025,    0.026026,    0.027027,    0.028028,    0.029029,     0.03003,    0.031031,    0.032032,
          0.048048,    0.049049,     0.05005,    0.051051,    0.052052,    0.053053,    0.054054,    0.055055,    0.056056,
          0.072072,    0.073073,    0.074074,    0.075075,    0.076076,    0.077077,    0.078078,    0.079079,     0.08008,
          0.096096,    0.097097,    0.098098,    0.099099,      0.1001,      0.1011,      0.1021,      0.1031,      0.1041,
           0.12012,     0.12112,     0.12212,     0.12312,     0.12412,     0.12513,     0.12613,     0.12713,     0.12813,
           0.14414,     0.14515,     0.14615,     0.14715,     0.14815,     0.14915,     0.15015,     0.15115,     0.15215,
           0.16817,     0.16917,     0.17017,     0.17117,     0.17217,     0.17317,     0.17417,     0.17518,     0.17618,
           0.19219,     0.19319,     0.19419,      0.1952,      0.1962,      0.1972,      0.1982,      0.1992,      0.2002,
           0.21622,     0.21722,     0.21822,     0.21922,     0.22022,     0.22122,     0.22222,     0.22322,     0.22422,
           0.24024,     0.24124,     0.24224,     0.24324,     0.24424,     0.24525,     0.24625,     0.24725,     0.24825,
           0.26426,     0.26527,     0.26627,     0.26727,     0.26827,     0.26927,     0.27027,     0.27127,     0.27227,
           0.28829,     0.28929,     0.29029,     0.29129,     0.29229,     0.29329,     0.29429,      0.2953,      0.2963,
           0.31231,     0.31331,     0.31431,     0.31532,     0.31632,     0.31732,     0.31832,     0.31932,     0.32032,
           0.33634,     0.33734,     0.33834,     0.33934,     0.34034,     0.34134,     0.34234,     0.34334,     0.34434,
           0.36036,     0.36136,     0.36236,     0.36336,     0.36436,     0.36537,     0.36637,     0.36737,     0.36837,
           0.38438,     0.38539,     0.38639,     0.38739,     0.38839,     0.38939,     0.39039,     0.39139,     0.39239,
           0.40841,     0.40941,     0.41041,     0.41141,     0.41241,     0.41341,     0.41441,     0.41542,     0.41642,
           0.43243,     0.43343,     0.43443,     0.43544,     0.43644,     0.43744,     0.43844,     0.43944,     0.44044,
           0.45646,     0.45746,     0.45846,     0.45946,     0.46046,     0.46146,     0.46246,     0.46346,     0.46446,
           0.48048,     0.48148,     0.48248,     0.48348,     0.48448,     0.48549,     0.48649,     0.48749,     0.48849,
            0.5045,     0.50551,     0.50651,     0.50751,     0.50851,     0.50951,     0.51051,     0.51151,     0.51251,
           0.52853,     0.52953,     0.53053,     0.53153,     0.53253,     0.53353,     0.53453,     0.53554,     0.53654,
           0.55255,     0.55355,     0.55455,     0.55556,     0.55656,     0.55756,     0.55856,     0.55956,     0.56056,
           0.57658,     0.57758,     0.57858,     0.57958,     0.58058,     0.58158,     0.58258,     0.58358,     0.58458,
            0.6006,      0.6016,      0.6026,      0.6036,      0.6046,     0.60561,     0.60661,     0.60761,     0.60861,
           0.62462,     0.62563,     0.62663,     0.62763,     0.62863,     0.62963,     0.63063,     0.63163,     0.63263,
           0.64865,     0.64965,     0.65065,     0.65165,     0.65265,     0.65365,     0.65465,     0.65566,     0.65666,
           0.67267,     0.67367,     0.67467,     0.67568,     0.67668,     0.67768,     0.67868,     0.67968,     0.68068,
            0.6967,      0.6977,      0.6987,      0.6997,      0.7007,      0.7017,      0.7027,      0.7037,      0.7047,
           0.72072,     0.72172,     0.72272,     0.72372,     0.72472,     0.72573,     0.72673,     0.72773,     0.72873,
           0.74474,     0.74575,     0.74675,     0.74775,     0.74875,     0.74975,     0.75075,     0.75175,     0.75275,
           0.76877,     0.76977,     0.77077,     0.77177,     0.77277,     0.77377,     0.77477,     0.77578,     0.77678,
           0.79279,     0.79379,     0.79479,      0.7958,      0.7968,      0.7978,      0.7988,      0.7998,      0.8008,
           0.81682,     0.81782,     0.81882,     0.81982,     0.82082,     0.82182,     0.82282,     0.82382,     0.82482,
           0.84084,     0.84184,     0.84284,     0.84384,     0.84484,     0.84585,     0.84685,     0.84785,     0.84885,
           0.86486,     0.86587,     0.86687,     0.86787,     0.86887,     0.86987,     0.87087,     0.87187,     0.87287,
           0.88889,     0.88989,     0.89089,     0.89189,     0.89289,     0.89389,     0.89489,      0.8959,      0.8969,
           0.91291,     0.91391,     0.91491,     0.91592,     0.91692,     0.91792,     0.91892,     0.91992,     0.92092,
           0.93694,     0.93794,     0.93894,     0.93994,     0.94094,     0.94194,     0.94294,     0.94394,     0.94494,
           0.96096,     0.96196,     0.96296,     0.96396,     0.96496,     0.96597,     0.96697,     0.96797,     0.96897,
           0.98498,     0.98599,     0.98699,     0.98799,     0.98899,     0.98999,     0.99099,     0.99199,     0.99299,
                  1,           1,           1,           1,           1,           1,           1,           1,           1,
```

```
model.predict(source="/content/test/images", conf=0.25, save=True)
```

```
image 23/67 /content/test/images/potholes275_png.rf.804630de6c1a2884604e5d87d5eb0ca0.jpg: 640x640 1 pothole, 148.0ms
image 24/67 /content/test/images/potholes283_png.rf.17094e0305fc42e33533103e7dcbdaa5.jpg: 640x640 2 potholes, 147.4ms
image 25/67 /content/test/images/potholes290_png.rf.399db13e76f745660c912e76a0092bed.jpg: 640x640 1 pothole, 147.0ms
image 26/67 /content/test/images/potholes296_png.rf.35ed863329afca114c2db6a41238ffa6.jpg: 640x640 1 pothole, 153.2ms
image 27/67 /content/test/images/potholes2_png.rf.bf7891a70a2a2b1d9038198f55de6d62.jpg: 640x640 3 potholes, 167.4ms
image 28/67 /content/test/images/potholes313_png.rf.268e443a3c33545c5bc1a43470e619aa.jpg: 640x640 1 pothole, 150.0ms
image 29/67 /content/test/images/potholes322_png.rf.81e60c2eef0d1ffc69fbb400302bea19.jpg: 640x640 2 potholes, 154.8ms
image 30/67 /content/test/images/potholes324_png.rf.67009da534f599be367040016322d54e.jpg: 640x640 1 pothole, 142.3ms
image 31/67 /content/test/images/potholes347_png.rf.8cdebc5fb1d491cf6b227a660a0e9ed1.jpg: 640x640 1 pothole, 143.2ms
```

```
image 55/67 /content/test/images/potholes548_png.rf.2c22a5f2074450ld46ac84a78479ed0b.jpg: 640x640 4 potholes, 143.1ms
image 56/67 /content/test/images/potholes54_png.rf.91f283529dede33c001980275c3633d0.jpg: 640x640 1 pothole, 148.3ms
image 57/67 /content/test/images/potholes555_png.rf.588f471ea45495bb36a8d6b6859b0837.jpg: 640x640 2 potholes, 144.9ms
image 58/67 /content/test/images/potholes589_png.rf.f483fc914dc5277a71990972284b5b89.jpg: 640x640 1 pothole, 152.6ms
image 59/67 /content/test/images/potholes605_png.rf.bc6675137d3e70241b23b92f1e757d62.jpg: 640x640 3 potholes, 155.8ms
image 60/67 /content/test/images/potholes608_png.rf.2c36f92a1463384a729d60d7dcb42494.jpg: 640x640 1 pothole, 152.7ms
image 61/67 /content/test/images/potholes613_png.rf.78ccfc4028752ffb3a33701c2b7b99c5.jpg: 640x640 3 potholes, 151.2ms
image 62/67 /content/test/images/potholes614_png.rf.9b1f1069fe95a4ced510f7f7d381ebb2.jpg: 640x640 1 pothole, 152.1ms
image 63/67 /content/test/images/potholes617_png.rf.c8ad5b3f3286b074ca1d575f10746f61.jpg: 640x640 3 potholes, 156.0ms
image 64/67 /content/test/images/potholes633_png.rf.562bd7109018d4bdc6bfe545daa637e8.jpg: 640x640 3 potholes, 148.7ms
image 65/67 /content/test/images/potholes659_png.rf.a8b4ea04f760889521019da7d82197ab.jpg: 640x640 3 potholes, 144.5ms
image 66/67 /content/test/images/potholes67_png.rf.396bf1ea40911cf5292e19a3cdd6780d.jpg: 640x640 2 potholes, 143.7ms
image 67/67 /content/test/images/potholes9_png.rf.02eafffef1cc447ed3f4072addd3e17d.jpg: 640x640 1 pothole, 153.0ms
Speed: 2.1ms preprocess, 153.4ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 640)
Results saved to /content/runs/detect/predict
[ultralytics.engine.results.Results object with attributes:

 boxes: ultralytics.engine.results.Boxes object
 keypoints: None
 masks: None
 names: {0: 'pothole'}
 obb: None
 orig_img: array([[[ 96,  82,  84],
         [ 94,  81,  83],
         [106,  93,  95],
```

```
model.val(data="/content/data.yaml", plots=True)
```

```
Ultralytics 8.4.16 🚀 Python-3.12.12 torch-2.10.0+cpu CPU (Intel Xeon CPU @ 2.20GHz)
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1703.0±420.4 MB/s, size: 81.7 KB)
val: Scanning /content/valid/labels.cache... 133 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 133/133 37.2Mit/s 0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 9/9 4.9s/it 44.0s
                   all        133        411      0.865      0.652       0.81      0.521
Speed: 7.9ms preprocess, 313.3ms inference, 0.0ms loss, 1.0ms postprocess per image
Results saved to /content/runs/detect/val2
ultralytics.utils.metrics.DetMetrics object with attributes:

ap_class_index: array([0])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at 0x7fac0f918b30>
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-Confidence(B)', 'Recall-Confidence(B)']
curves_results: [[array([          0,    0.001001,    0.002002,    0.003003,    0.004004,    0.005005,    0.006006,
0.007007,    0.008008,    0.009009,     0.01001,    0.011011,    0.012012,    0.013013,    0.014014,    0.015015,    0.016016,
0.017017,    0.018018,    0.019019,     0.02002,    0.021021,    0.022022,    0.023023,
           0.024024,    0.025025,    0.026026,    0.027027,    0.028028,    0.029029,     0.03003,    0.031031,    0.032032,
0.033033,    0.034034,    0.035035,    0.036036,    0.037037,    0.038038,    0.039039,     0.04004,    0.041041,    0.042042,
0.043043,    0.044044,    0.045045,    0.046046,    0.047047,
           0.048048,    0.049049,     0.05005,    0.051051,    0.052052,    0.053053,    0.054054,    0.055055,    0.056056,
0.057057,    0.058058,    0.059059,     0.06006,    0.061061,    0.062062,    0.063063,    0.064064,    0.065065,    0.066066,
0.067067,    0.068068,    0.069069,     0.07007,    0.071071,
           0.072072,    0.073073,    0.074074,    0.075075,    0.076076,    0.077077,    0.078078,    0.079079,     0.08008,
0.081081,    0.082082,    0.083083,    0.084084,    0.085085,    0.086086,    0.087087,    0.088088,    0.089089,     0.09009,
0.091091,    0.092092,    0.093093,    0.094094,    0.095095,
           0.096096,    0.097097,    0.098098,    0.099099,      0.1001,      0.1011,      0.1021,      0.1031,      0.1041,
0.10511,     0.10611,     0.10711,     0.10811,     0.10911,     0.11011,     0.11111,     0.11211,     0.11311,     0.11411,
0.11512,     0.11612,     0.11712,     0.11812,     0.11912,
            0.12012,     0.12112,     0.12212,     0.12312,     0.12412,     0.12513,     0.12613,     0.12713,     0.12813,
0.12913,     0.13013,     0.13113,     0.13213,     0.13313,     0.13413,     0.13514,     0.13614,     0.13714,     0.13814,
0.13914,     0.14014,     0.14114,     0.14214,     0.14314,
            0.14414,     0.14515,     0.14615,     0.14715,     0.14815,     0.14915,     0.15015,     0.15115,     0.15215,
0.15315,     0.15415,     0.15516,     0.15616,     0.15716,     0.15816,     0.15916,     0.16016,     0.16116,     0.16216,
0.16316,     0.16416,     0.16517,     0.16617,     0.16717,
            0.16817,     0.16917,     0.17017,     0.17117,     0.17217,     0.17317,     0.17417,     0.17518,     0.17618,
0.17718,     0.17818,     0.17918,     0.18018,     0.18118,     0.18218,     0.18318,     0.18418,     0.18519,     0.18619,
0.18719,     0.18819,     0.18919,     0.19019,     0.19119,
            0.19219,     0.19319,     0.19419,      0.1952,      0.1962,      0.1972,      0.1982,      0.1992,      0.2002,
0.2012,      0.2022,      0.2032,      0.2042,     0.20521,     0.20621,     0.20721,     0.20821,     0.20921,     0.21021,
0.21121,     0.21221,     0.21321,     0.21421,     0.21522,
            0.21622,     0.21722,     0.21822,     0.21922,     0.22022,     0.22122,     0.22222,     0.22322,     0.22422,
0.22523,     0.22623,     0.22723,     0.22823,     0.22923,     0.23023,     0.23123,     0.23223,     0.23323,     0.23423,
0.23524,     0.23624,     0.23724,     0.23824,     0.23924,
            0.24024,     0.24124,     0.24224,     0.24324,     0.24424,     0.24525,     0.24625,     0.24725,     0.24825,
0.24925,     0.25025,     0.25125,     0.25225,     0.25325,     0.25425,     0.25526,     0.25626,     0.25726,     0.25826,
0.25926,     0.26026,     0.26126,     0.26226,     0.26326,
            0.26426,     0.26527,     0.26627,     0.26727,     0.26827,     0.26927,     0.27027,     0.27127,     0.27227,
0.27327,     0.27427,     0.27528,     0.27628,     0.27728,     0.27828,     0.27928,     0.28028,     0.28128,     0.28228,
0.28328,     0.28428,     0.28529,     0.28629,     0.28729,
            0.28829,     0.28929,     0.29029,     0.29129,     0.29229,     0.29329,     0.29429,      0.2953,      0.2963,
0.2973,      0.2983,      0.2993,      0.3003,      0.3013,      0.3023,      0.3033,      0.3043,     0.30531,     0.30631,
0.30731,     0.30831,     0.30931,     0.31031,     0.31131,
            0.31231,     0.31331,     0.31431,     0.31532,     0.31632,     0.31732,     0.31832,     0.31932,     0.32032,
0.32132,     0.32232,     0.32332,     0.32432,     0.32533,     0.32633,     0.32733,     0.32833,     0.32933,     0.33033,
0.33133,     0.33233,     0.33333,     0.33433,     0.33534,
            0.33634,     0.33734,     0.33834,     0.33934,     0.34034,     0.34134,     0.34234,     0.34334,     0.34434,
```

```
0.34535,     0.34635,     0.34735,     0.34835,     0.34935,     0.35035,     0.35135,     0.35235,     0.35335,     0.35435,
0.35536,     0.35636,     0.35736,     0.35836,     0.35936,
```

```
print("Precision:", metrics.results_dict['metrics/precision(B)'])
print("Recall:", metrics.results_dict['metrics/recall(B)'])
print("mAP50:", metrics.results_dict['metrics/mAP50(B)'])
print("mAP50-95:", metrics.results_dict['metrics/mAP50-95(B)'])
```

```
Precision: 0.864542914350951
Recall: 0.6520681265206812
mAP50: 0.8102675380825334
mAP50-95: 0.5205572266711194
```

Start coding or generate with AI.